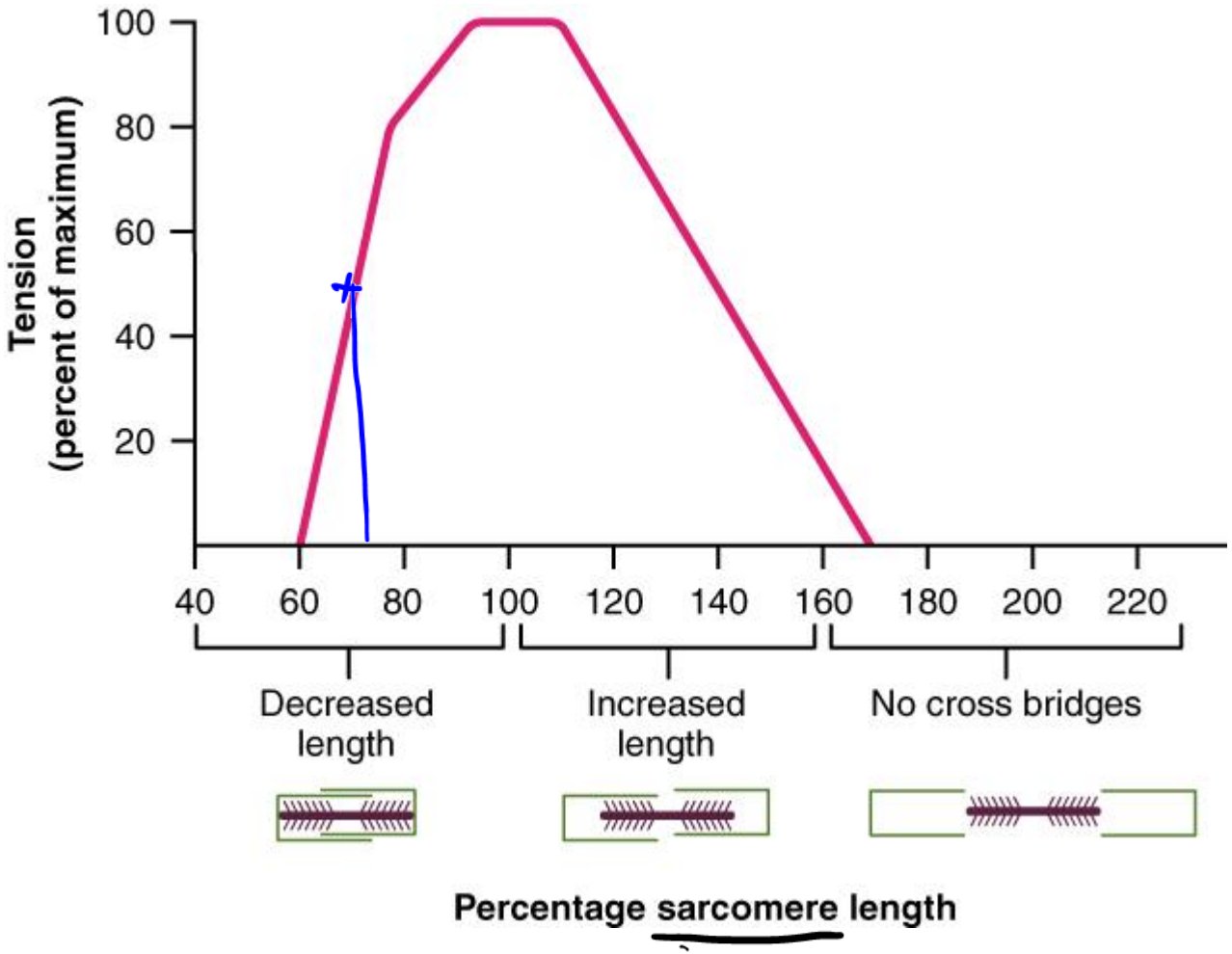
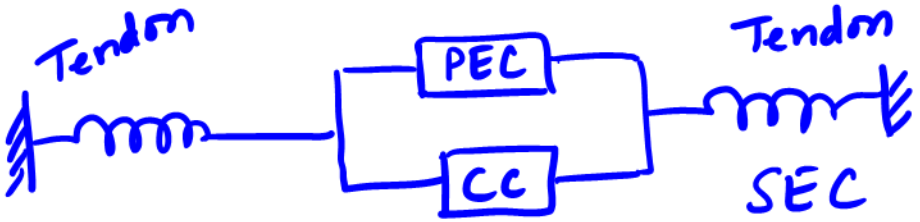
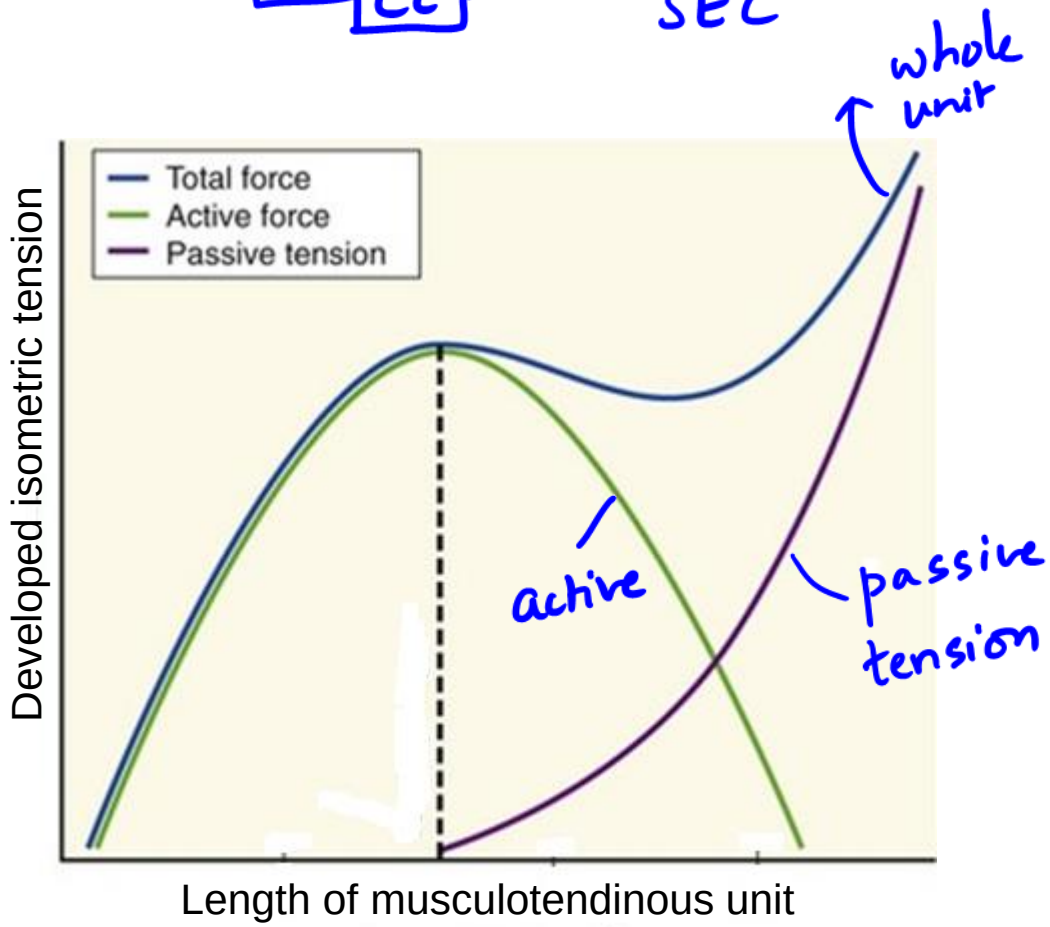
Length-Tension relationship



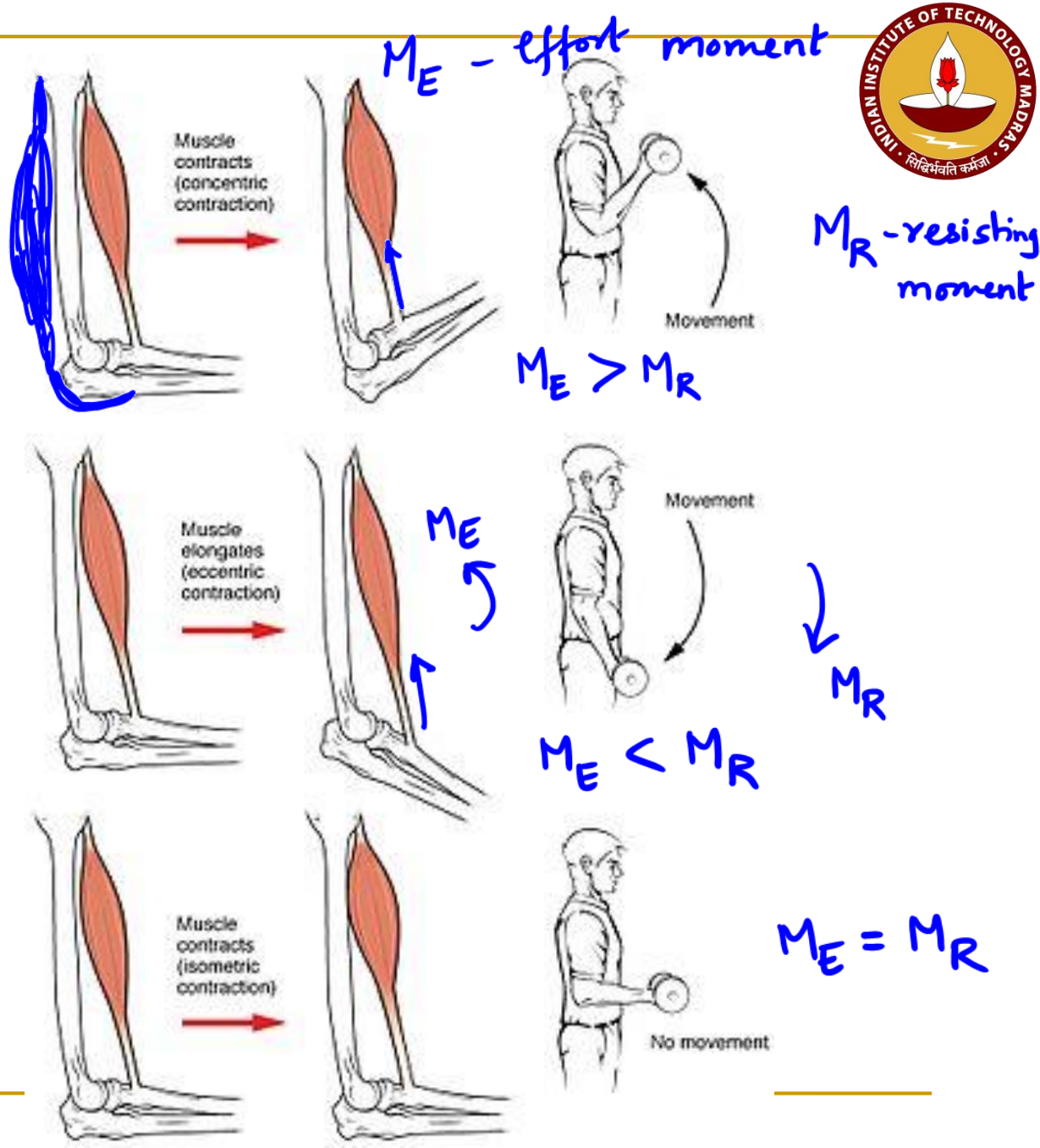
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Muscle action

- Concentric action – muscle shortens
- Eccentric action – muscle lengthens
- Isometric action – length remains the same



TTK Center for Rehabilitation Research & Device Development (R2D2)
Department of Mechanical Engineering
https://home.iitm.ac.in/r2d2

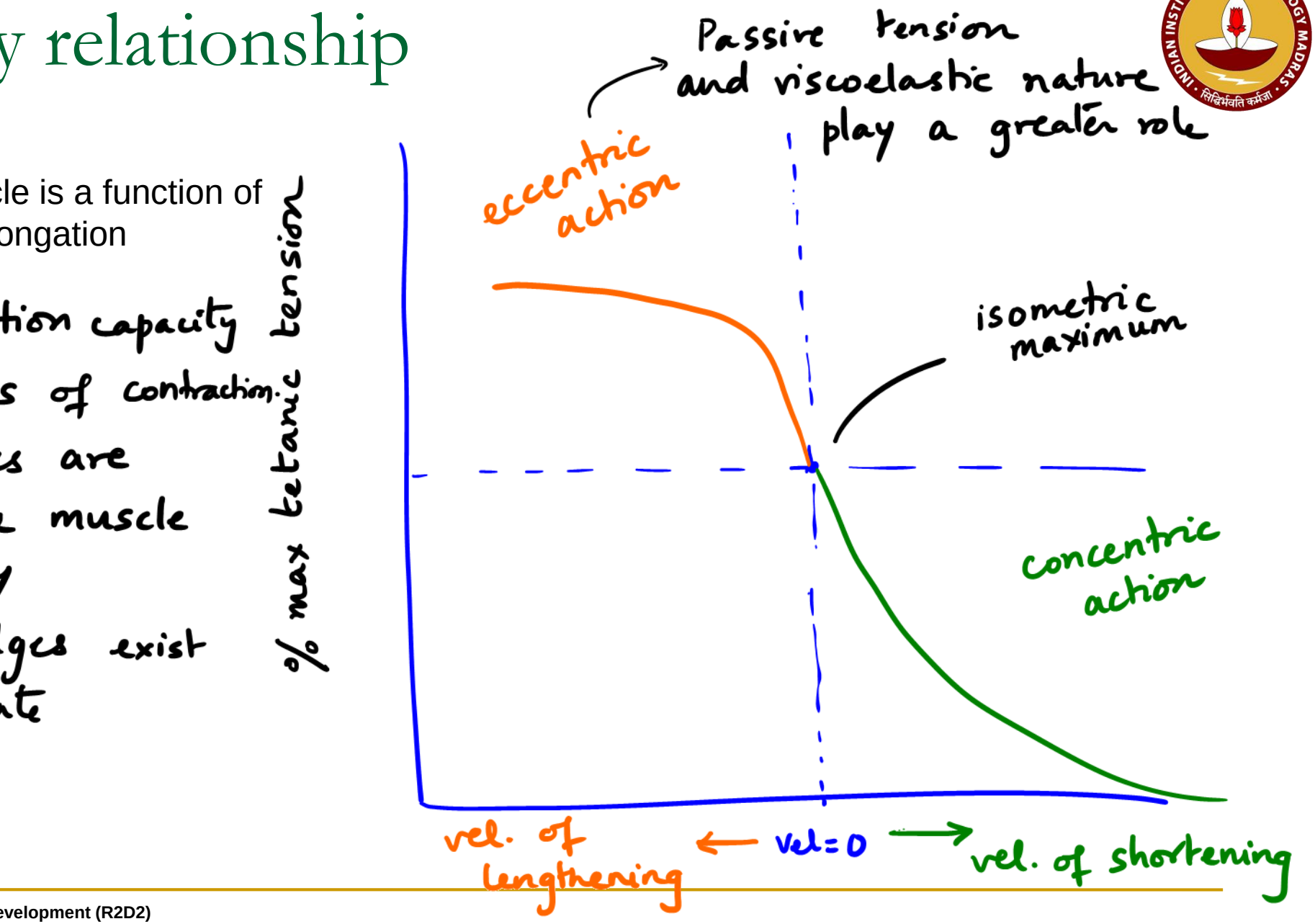
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Force-velocity relationship

Force generated by a muscle is a function of its velocity of contraction/elongation

Reduced force generation capacity at higher velocities of contraction.
∵ no. of cross bridges are fewer than when the muscle is contracting slowly

Max. no. of cross bridges exist in the isometric state



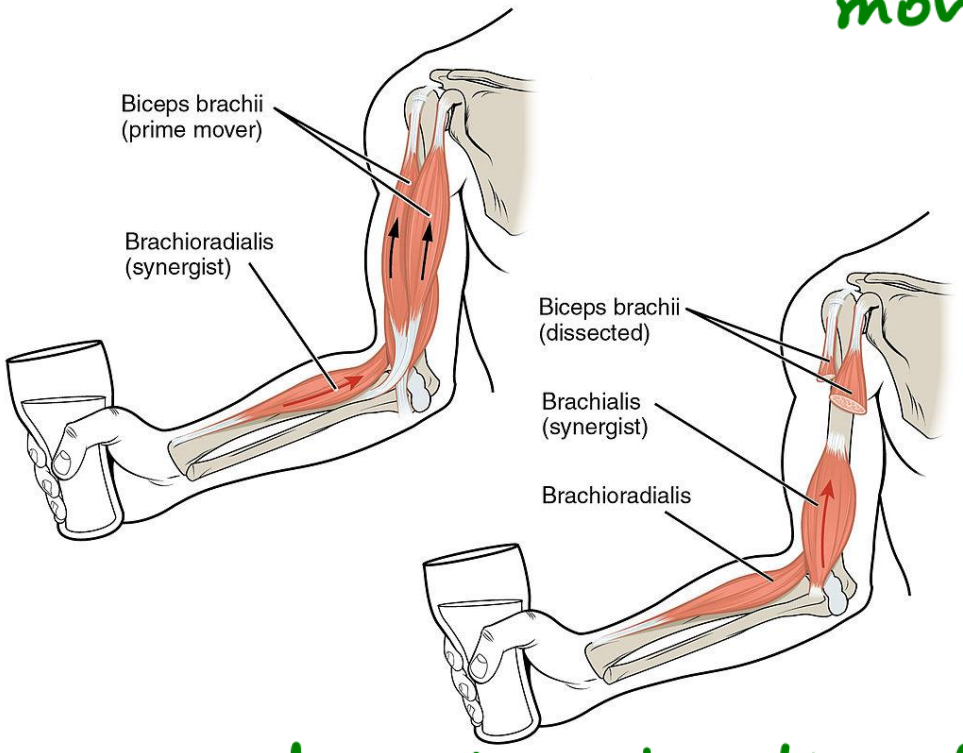
Muscle action

- Agonists
- Antagonists
- Stabilizers
- Synergists or Neutralizers

shoulder girdle
pelvic girdle

Stabilizers : act in one segment so that a specific movement in an adjacent joint can occur. Undergo minimal movement





Neutralizer : muscle contracts to eliminate an undesired joint action of another muscle

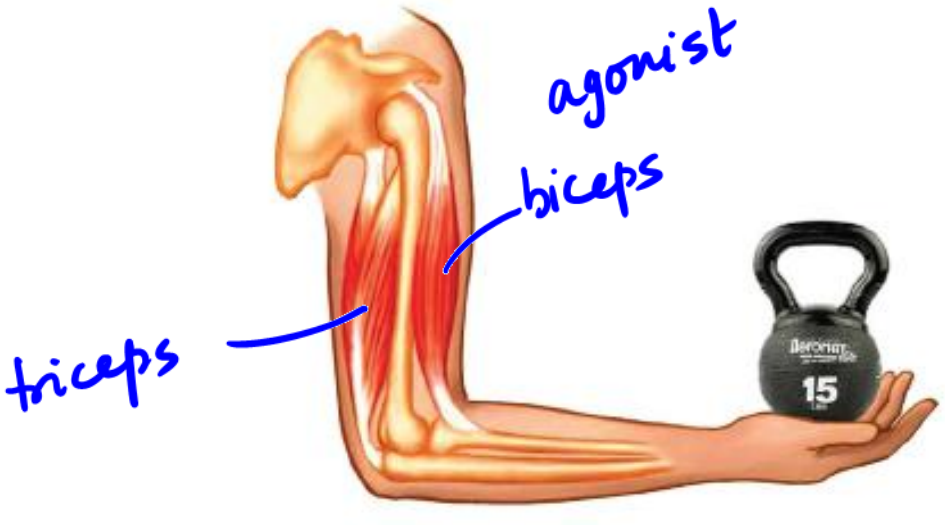
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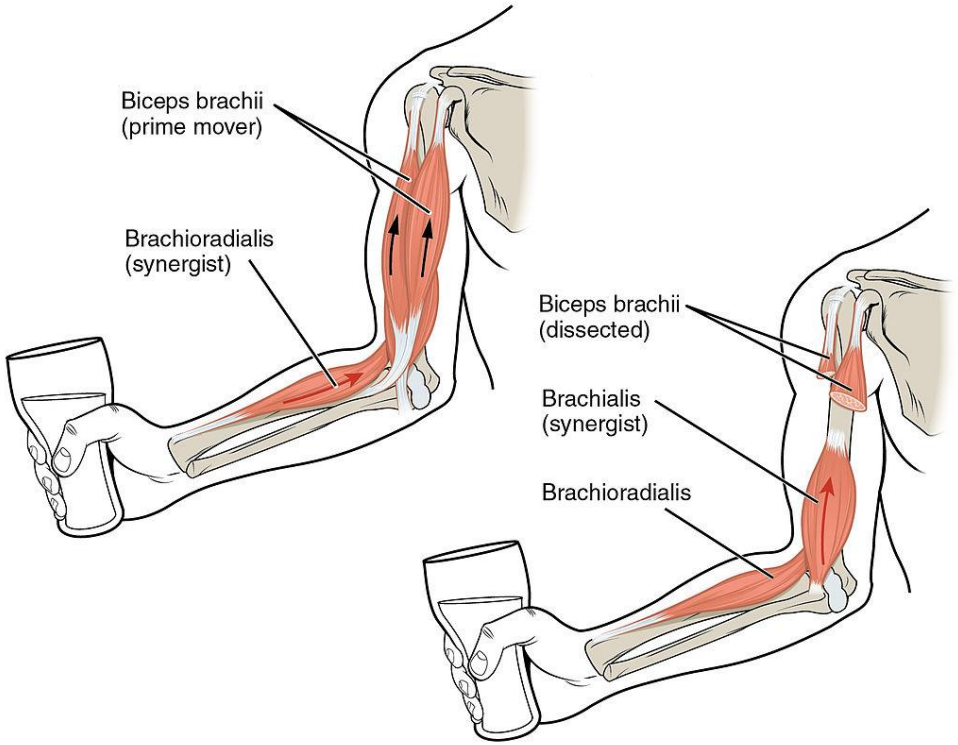
Muscle action

Agonists - muscles creating the same joint movement

Antagonists - muscles opposing or producing the opposite joint movement

- Agonists
- Antagonists
- Stabilizers
- Synergists or Neutralizers





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